

# ZHENG QING

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## Education

### Purdue University

Expected May 2027

*B.S. in Electrical Engineering | Cumulative GPA 3.83/4.0 | Major GPA 3.95/4.0*

*West Lafayette, IN*

- **Relevant coursework:** Linear Control Systems, Signals & Systems, Probabilistic Methods, Digital System Design, Microprocessor Systems, Analog Circuits.

## Research Experience

### Senior Design — 3D Vehicle State Estimation and NMPC Trajectory Tracking

Sept 2026 – Present

*Advised by Prof. Ziran Wang, Purdue Digital Twin Lab*

*West Lafayette, IN*

- Porting the TUM FTM 3D vehicle dynamics state estimator — a 3D point-mass EKF supported by a single-track UKF — from a race-car platform onto a research vehicle with a weaker sensor set, in **C++** on **ROS 2 Humble**.
- Implementing an **NMPC** trajectory-tracking controller on top of the estimator so that the effect of estimation quality on closed-loop tracking error can be measured rather than assumed.
- Characterizing which state component — longitudinal velocity, yaw rate, or sideslip angle — dominates tracking degradation when GNSS drops out.

### UAV Ortho-Mosaic Reconstruction under an Unreliable Pose Model

Jul 2026 – Present

*Research Assistant, Purdue Airlab*

*West Lafayette, IN*

- Built a **Python/OpenCV** stitching pipeline compositing **1,145 drone frames across 11 flight strips** into a 99-megapixel ortho-mosaic, recovering camera poses by **global bundle adjustment** after onboard heading and GNSS proved untrustworthy as pose ground truth.
- Isolated the binding constraint by benchmarking against a registered reference: texture survives stitching essentially losslessly, so residual error is attributable to a pose model carrying **no per-frame heading and no scale**.
- Showed that seam artifacts scale with the **baseline** between contributing frames rather than the time gap, redirecting the fix from exposure/blending toward geometry.

### Purdue VIP ARES (Autonomous Race-Engineering Systems), Control Team

Jan 2026 – May 2026

*Undergraduate Researcher*

*West Lafayette, IN*

- Developed the low-level controller through a full progression — **PID** → **ADRC** → **LQR** → **MPC** — moving to optimal control once the objective had to trade tracking error, control effort, and stability against strategy-dependent priorities.
- Contributed to a hierarchical **strategy / planning / execution** architecture connecting high-level team objectives to real-time multi-agent control.
- Built a modular **ROS 2** evaluation pipeline — topic-level control, odometry state feedback, and time-synchronized logging — and ran comparative analysis of control strategies on a simulated Shanghai circuit.
- Co-authored the resulting team technical report and the companion conference poster.

### Quantum Optics Laboratory

May 2025 – Oct 2025

*SURF Research Fellow, mentored by Prof. Shengwang Du | \$6,500 fellowship*

*West Lafayette, IN*

- Designed and built a current-control module for a Magneto-Optical Trap end to end: **PCB design**, thermal management in a fan-cooled metal enclosure, and circuit validation with Hall-probe feedback.
- Presented a poster on biphoton generation and the MOT current-control box at the SURF Summer Symposium.

## Poster Publications & Presentations

### A Strategy-Driven Multi-Agent Control Framework for Autonomous Racing

Spring 2026

- Z. Wang, K. Zhu, **Z. Qing**, K. V. Desiredi, J. Zhang, B. Lee. Purdue VIP ARES Control Team group technical report (not peer-reviewed).

**Controller Design for an Autonomous Racing Vehicle** — poster, CCAT Global Symposium on Mobility Innovation, University

- **ITS Michigan Scholar Award** — undergraduate winner, student poster competition.

## Additional Experience & Projects

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### Undergraduate Teaching Assistant — Elmore Family School of ECE

May 2026 – Present

Purdue University

West Lafayette, IN

- **ECE 20875 — Python for Data Science** (Fall 2026): core EE course covering Python data structures and higher-order functions alongside probability distributions, sampling and hypothesis testing, regression with regularization and cross-validation, clustering, and classification.
- **ECE 20001 — Electrical Engineering Fundamentals I** (Summer 2026): supported a cohort of ~300 students in Purdue's introductory circuits course — DC and AC circuit analysis, first-order transients, and diode and transistor fundamentals.

### Purdue SoCET VIP — System-on-Chip Extension Technologies

Aug 2024 – May 2025

Research Student, Circuit Design Team

West Lafayette, IN

- SRAM and clock design: circuit design in **LTspice** and **Cadence Virtuoso**, SystemVerilog FSMs, **KiCad** PCB layout, and RISC-V embedded work in C with peripheral drivers.
- Presented at the Spring Undergraduate Research Symposium to an audience of over 100.

### Flairmicro — Digital IC Design & Packaging

Aug 2023 – Sept 2023

Engineering Intern

Fuzhou, China

- Worked on Bluetooth communication modules for automotive applications; assisted with digital IC packaging, layout optimization, and production-line ESD compliance.

### VGA Pong on FPGA (*SystemVerilog, Pico-ICE FPGA*) — ECE 270 team project

Mar 2025 – May 2025

- Designed and verified the VGA timing generator ( $640 \times 480$  @ 60 Hz) and the pixel rendering module; implemented FSM-based game logic and collision detection through simulation, synthesis, and hardware debugging.

**Honors** — Honorable Mention, Mathematical Contest in Modeling (MCM/ICM); Semester Honors and Dean's List, Purdue University

## Technical Skills

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**Programming/HDL:** C++, Python, C, MATLAB, SystemVerilog

**Robotics & Control:** ROS 2 (Humble), EKF/UKF, NMPC, LQR, trajectory tracking, sensor fusion, bundle adjustment, OpenCV

**EDA/Tools:** Cadence Virtuoso, LTspice, KiCad  $\mu$ Vision, Git, Docker, VS Code

**Platforms/Systems:** Linux

**Spoken Languages:** Mandarin (native), English (fluent)

## Leadership & Extracurricular

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### PUCSSA — Chinese Students and Scholars Association at Purdue

Sept 2025 – Present

Minister, Career Development

West Lafayette, IN

- Organize career development events: resume workshops, mock interviews, and alumni networking sessions. Collaborate with companies like Midea, Eli Lilly, Huaqin

### Music

Sept 2021 – Dec 2023

President, High School Music Club; Lead Guitarist, One Tap Band

Suzhou, China

- Lead guitarist in a local band performing British rock, hard rock, and punk metal; coordinated ensemble rehearsals and mentored junior members.